Introduction

Cardiovascular Risk Associated with Autoimmune Rheumatic Disease (ARD) Patient Treatment in Dubai, United Arab Emirates

Autoimmune disease is defined as a condition in which the body's immune system attacks healthy cells (Watson, 2019). When the immune system kills these cells, a buildup of these cells, called inflammation, occurs (Watson, 2019). Widespread and persistent inflammation can cause damage to arteries, joints, and organs (Watson, 2019). In Autoimmune Rheumatic Diseases (ARD), immune cells target joints, tendons, cartilage, and internal organs in the body, and the resulting inflammation can cause significant pain, swelling, impairment of function, and eventually, if untreated, may cause permanent disability (Johns Hopkins Rheumatology, 2018; Quest Diagnostics, 2019). ARD cases are on the rise worldwide, but whether this is due to an increase in true incidence or an increase in awareness is unclear (Namas et. al, 2019). While the cause of ARD is unknown, many risk factors such as smoking, genetic predisposition, and hormonal imbalances have been documented (Nagy et. al, 2018), and these conditions can affect patients of all age groups, including children (Watson, 2019). A decade ago, due to lack of awareness of these conditions among the general population, there was a significant delay in diagnosis. Due to a number of initiatives, there appears to be a reduction in the delay of diagnosis which reflects an increase in incidence and prevalence of ARDs in UAE (Namas et al.,2020). According to treatment recommendations, most of these patients require an initial treatment with a high dosage of corticosteroids, a group of anti-inflammatory drugs, to induce remission (DeMizio & Geraldino-Pardilla, 2019). While inflammation decreases with the use of corticosteroids, many complications during and after administration of these drugs have caused an increased cardiovascular risk in patients (Alhusain & Bruce, 2013; DeMizio & Geraldino-Pardilla, 2019; Hannawi et. al, 2019; Mahtta et. al, 2020). While many speculate that prolonged corticosteroid treatment is contributing to this increased risk (Alhusain & Bruce, 2013; DeMizio & Geraldino-Pardilla, 2019), no definitive data in the United Arab Emirates (UAE) has been published. This research aims to increase the understanding of impacts of certain risk factors and their impact on CVD risk in the Middle East and further the relevancy of (swarat's paper) through expanding scope of study of risks.

Definition of Terms

Terminology	Definition
Autoimmune Disease (AD)	Autoimmune Disease is defined as a condition in which the body's immune system attacks healthy cells (Watson, 2019)
Autoimmune Rheumatic Diseases (ARD)	Autoimmune Rheumatic Diseases are ADs where immune cells target joints, tendons,

	cartilage, and internal organs in the body (Johns Hopkins Rheumatology, 2018; Quest Diagnostics, 2019).
United Arab Emirates (UAE)	The United Arab Emirates is the Middle Eastern region in which the study is conducted. The UAE is unique due to its Eastern ethnic diversity and its support for alternative medicine (D'Mello, 2019; Namas et. al, 2019; University of Birmingham, 2021).
Diabetes Mellitus (DM)	Diabetes Mellitus are a group of diseases that result in too much sugar (glucose) in the blood. DM has been proven to increase the risk of cardiovascular disease (Durante & Bronzato, 2015; NHS UK, 2018).
Hypertension (HTN)	Hypertension, or high blood pressure is a condition in which the long-term force of the blood against your artery walls is high enough that it may eventually cause health problems, such as cardiovascular disease (Durante & Bronzato, 2015; NHS UK, 2018).
Cardiovascular Disease	Cardiovascular Disease is defined as the group of diseases that affect the circulatory system, including the heart and its constituent vessels (American Heart Association, 2021).
Cardiovascular Risk	Cardiovascular risk is defined as the probability of having some form of cardiovascular disease in the patient's lifetime (American Heart Association, 2021). Cardiovascular risk fluctuates with many internal and external factors (Durante & Bronzato, 2015).
Corticosteroids	Corticosteroids, also known as steroids, are a group of anti-inflammatory medications used as a first line of treatment for ARD patients (Hall, 1999; Westphalen, 2019). They have many side effects, including adrenal dependency and increasing the risk

	of DM and HTN (Broersen et. al, 2015; Mayo Clinic Staff, 2020).
Disease Modifying Anti-Rheumatic Drugs (DMARDs)	Disease Modifying Anti-Rheumatic Drugs are a type of NSAID that targets specific immune cells that attack healthy joints to more effectively reduce disease symptoms (Cohen & Cannella, 2019). One type of DMARD, biologics (DeMizio & Geraldino-Pardilla, 2019), use molecules naturally produced in the body to mitigate side effects.
Tumor Necrosis Factor (TNF) Inhibitors	Tumor Necrosis Factor (TNF) Inhibitors are a type of biologic that suppresses the physiologic response to tumor necrosis factor, which is part of the inflammatory response.
Nonbiologic Disease Modifying Anti-Rheumatic Drugs (DMARDs)	Nonbiologic Disease Modifying Anti-Rheumatic Drugs (DMARDs) are a type of DMARD that suppresses the inflammatory response and blocks the proliferation of rapidly dividing cells
Symptoms	Symptoms refer to the date that joint symptoms first present in ARD study patients (Dubai Health Authority, 2021).
Diagnosis	Diagnosis refers to the date of first screening of the ARD study patients by any doctor (Dubai Health Authority, 2021).
Treatment	Treatment refers to the date of the first check-up by a rheumatologist (Dubai Health Authority, 2021).

Applied Sub-Problem: Biologics and Steroids and HTN

HTN is one of the risk factors that contribute to CVD. In order to reduce the risk of CVD in patients with RA, patient medication data is analyzed to further our understanding of how certain drug classes affect CVD risk association. The drugs considered in this sub-problem are Biologics, medication/therapies that are made from microorganisms, and Steroids, medication containing natural/synthetic hormones. Since we already know that patients of Middle Eastern ethnicities are more likely to develop symptoms of CVD risk, it is important to further our understanding of drug classes and their effects on CVD risk association to better their treatment and reduce their chances of CVD risk.

Methods

To determine if there is a statistically lower chance of the probability of having HTN while using biologics versus steroids, raw data was first filtered to determine the amount of patients using a type of biologic with the comorbidity of HTN, and then was filtered to determine the amount of patients using a steroidal treatment with the comorbidity of HTN.
Then a Prop 2 Z test was conducted with these statistics to determine if there is a lowered probability of HTN when using biologics versus using steroids. In both of these tests, $\alpha = 0.05$, and the data is considered random through sampling, independent because the sample is less than 10% of UAE population, and normal since the sample size is greater than 30 patients (Maverick, 2021). If the p-value of these tests are less than 0.05, the null hypothesis is rejected, indicating that there is a statistically significant probability of patients using a specific type of medicine to decrease risk of that comorbidity. If the p-value of these tests are greater than 0.05, the null fails to be rejected, indicating that there is no statistically significant probability of patients using a specific type of medicine to decrease risk of that comorbidity.

Results

With the retrospective data set, the ARD patient data was first filtered to determine the number of patients using biologics and steroids with HTN. Out of the 981 patients in this study, the data showed that the amount of patients using biologics with HTN is 22.23%, and the amount of patients using steroids with HTN is 51.02%. Since the *p* value of the test is less than $\alpha = 0.05$, the test concludes that the null hypothesis can be failed to be rejected; this indicates that using biologics versus steroids decreases the probability of HTN prevalence. This is consistent with current literature that states that biologics are helpful in decreasing HTN prevalence. In Morgan West Stewart rodriguez morhartu, patients using steroids have a 17% greater chance of HTN. Literature mainly points to biologics effectively controlling and treating inflammation, decreasing the risk of HTN in ARD patients.

Applied Sub-Problem : Biologics and Steroids and DM

DM is one of the risk factors that contribute to CVD. In order to reduce the risk of CVD in patients with RA, patient medication data is analyzed to further our understanding of how certain drug classes affect CVD risk association. The drugs considered in this sub-problem are Biologics, medication/therapies that are made from microorganisms, and Steroids, medication containing natural/synthetic hormones. Since we already know that patients of Middle Eastern ethnicities are more likely to develop symptoms of CVD risk, it is important to further our understanding of drug classes and their effects on CVD risk association to better their treatment and reduce their chances of CVD risk.

Methods

To determine if there is a statistically lowering chance of having DM while using biologics or steroids, the raw data was first filtered to determine the amount of patients using steroids with the comorbidity of DM, and then the amount of patients using biologics with the comorbidity of DM.

Then a Prop 2 Z test was conducted with these statistics to determine if there is a lowered probability of DMwhen using biologics versus using steroids. In both of these tests, $\alpha = 0.05$, and the data is considered random through sampling, independent because the sample is less than 10% of UAE population, and normal since the sample size is greater than 30 patients (Maverick, 2021). If the *p*-value of these tests are less than 0.05, the null hypothesis is rejected, indicating that there is a statistically significant probability of patients using a specific type of medicine to decrease risk of that comorbidity. If the *p*-value of these tests are greater than 0.05,

the null fails to be rejected, indicating that there is no statistically significant probability of patients using a specific type of medicine to decrease risk of that comorbidity.

Results

With the retrospective data set, the ARD patient data was first filtered to determine the number of patients using biologics and steroids withDM. Out of the 981 patients in this study, the data showed that the amount of patients using biologics with DM is 11.11%, and the amount of patients using steroids with DM is 10.11%. Since the p value of the test is greater than a = 0.05, the test concludes that the null hypothesis has failed to be rejected; this indicates that using biologics versus steroids does not decrease the probability of DM prevalence. While certain literature supports the fact that biologics can help effectively manage DM, most point to the fact that effects of biologics largely vary (balancing benefits and risks in the era of biologics, immunosuppressive treatment and the risk of diabetes in rheumatoid arthritis). Thus, our data can be explained by literature inconsistency on the effects of biologics.

Applied Sub-Problem: TNF Inhibitors and Steroids and DM

DM is one of the risk factors that contribute to CVD. In order to reduce the risk of CVD in patients with RA, patient medication data is analyzed to further our understanding of how certain drug classes affect CVD risk association. The drugs considered in this sub-problem are TNF inhibitors, a type of biologic that suppresses the physiologic response to tumor necrosis factor, which is part of the inflammatory response, and Steroids, medication containing natural/synthetic hormones. Since we already know that patients of Middle Eastern ethnicities are more likely to develop symptoms of CVD risk, it is important to further our understanding of drug classes and their effects on CVD risk association to better their treatment and reduce their chances of CVD risk.

Methods

To determine if there is a statistically lowering chance of having DM while using TNF inhibitors or steroids, the raw data was first filtered to determine the amount of patients using steroids with the co-morbidity of DM, and then the amount of patients using TNF inhibitors with the comorbidity of DM.
Then a 2-ProportionZ test was conducted with these statistics to determine if there is a lowered probability of DM when using biologics versus using steroids. In both of these tests, $\alpha = 0.05$, and the data is considered random through sampling, independent because the sample is less than 10% of UAE population, and normal since the sample size is greater than 30 patients (Maverick, 2021). If the p-value of these tests are less than 0.05, the null hypothesis is rejected, indicating that there is a statistically significant probability of patients using a specific type of medicine to decrease risk of that comorbidity. If the p-value of these tests are greater than 0.05,

the null fails to be rejected, indicating that there is no statistically significant probability of patients using a specific type of medicine to decrease risk of that comorbidity.

Results

With the retrospective data set, the ARD patient data was first filtered to determine the number of patients using TNF inhibitors and steroids with DM. Out of the 981 patients in this study, the data showed that the amount of patients using TNF inhibitors with DM is 9.09%, and the amount of patients using steroids with DM is 10.11%. Since the p value of the test is greater than a = 0.05, the test concludes that the null hypothesis has failed to be rejected; this indicates that using TNF inhibitors versus steroids does not decrease the probability of DM prevalence. This can be explained by the fact that the amount of people using TNF inhibitors compared to the amount of people using steroids is much lesser. Thus, the data might be more accurate if more people using TNF inhibitors were sampled, as literature points to TNF inhibitors effectively managing DM. Previous studies such as (JAMA network) have already significantly proved that TNF inhibitors manage DM more effectively compared to other groups of DMARDs. With the assumption that our results are consistent with literature, increasing the population size of TNF inhibitors might lead to more significant results.

Applied Sub-Problem: Non-biologic DMARDs and TNF inhibitors and DM

DM is one of the risk factors that contribute to CVD. In order to reduce the risk of CVD in patients with RA, patient medication data is analyzed to further our understanding of how certain drug classes affect CVD risk association. The drugs considered in this sub-problem are TNF inhibitors, a type of biologic that suppresses the physiologic response to tumor necrosis factor, which is part of the inflammatory response, and non-biologic DMARDs, that targets specific immune cells that attack healthy joints to more effectively reduce disease symptoms. Since we already know that patients of Middle Eastern ethnicities are more likely to develop symptoms of CVD risk, it is important to further our understanding of drug classes and their effects on CVD risk association to better their treatment and reduce their chances of CVD risk.

Methods

To determine if there is a statistically lowering chance of having DM while using TNF inhibitors or non-biologic DMARDs, the raw data was first filtered to determine the amount of patients using non-biologic DMARDs with the comorbidity of DM, and then the amount of patients using TNF inhibitors with the comorbidity of DM.
Then a Prop 2 Z test was conducted with these statistics to determine if there is a lowered probability of DM when using non-biologic DMARDs versus using steroids. In both of these tests, $\alpha = 0.05$, and the data is considered random through sampling, independent because the

sample is less than 10% of UAE population, and normal since the sample size is greater than 30 patients (Maverick, 2021). If the p-value of these tests are less than 0.05, the null hypothesis is rejected, indicating that there is a statistically significant probability of patients using a specific type of medicine to decrease risk of that comorbidity. If the p-value of these tests are greater than 0.05, the null fails to be rejected, indicating that there is no statistically significant probability of patients using a specific type of medicine to decrease risk of that comorbidity.

Results

With the retrospective data set, the ARD patient data was first filtered to determine the number of patients using TNF inhibitors and non-biologic DMARDs with DM. Out of the 981 patients in this study, the data showed that the amount of patients using TNF inhibitors with DM is 9.09%, and the amount of patients using non-biologic DMARDs with DM is 1.02%. Since the p value of the test is greater than a = 0.05, the test concludes that the null hypothesis has failed to be rejected; this indicates that using TNF inhibitors versus non-biologic DMARDs does not decrease the probability of DM prevalence. Only 1 person was using non-biologic DMARDs with DM, while 5 were using TNF inhibitors with DM. This makes the n<10, and therefore extremely weak. A stronger correlation might be found if future studies have a larger cohort of patients taking non-biologic DMARDs, as a greater n value may yield better results.

Conclusions

Literature indicates an increase in cardiovascular incidence for ARD patients worldwide (Alhusain & Bruce, 2013), but whether this is due to an increase in true incidence or an increase awareness is unclear (Namas et. al, 2019). While studies loosely correlate the effects of certain types of medication/treatment in cardiovascular risk factors, none show this correlation in Middle Eastern ARD patients. Therefore, this study looked specifically at the correlation between the factors specific medicines taken and the prevalence of CVD risk in Middle Eastern ARD patients. Understanding cardiovascular risk over incidence intends to bring researchers around the world a better picture of the underlying causes for this increase in cardiovascular cases among ARD patients.

Implications and Future Studies

Findings from this study firstly indicate that using biologics rather than steroids decreases a patient’s probability of HTN. This is supported by previous literature that has also found a

correlation between biologics and a lesser chance of HTN. As a result, more widespread usage of biologics in ARD treatment may cause a decrease in the patient's chance of HTN/cardiovascular risk factors. Notably, the relationship between biologics, steroids, and DM did not produce any significant results. However, this is mainly due to the unknown/unexplored nature of biologics and the effects it has on certain risk factors. Thus, future studies should take this into account in order to gain a better picture of biologics and their effects on DM. In addition, future studies should increase the cohorts of TNF inhibitors and non-biologic DMARDs in their dataset in order to yield better findings on their effect on DM, as the low n value of both these treatment types may have contributed to skewed findings in this study. Overall, while this paper did not produce many significant results, this research opens up many more questions to be answered for the benefit of Middle Eastern medicine. Tweaking study design slightly will surely generate more fruitful results in the future for characterizing cardiovascular risk in ARD patients worldwide.